Pascal's Triangle

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Outline

Pascal's Triangle

Symmetries

Row Sums

Binomial Theorem

Combinations

Question

There are n students. What is the number of ways of forming a team of k students out of them?

Combinations

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Answer

$$\binom{n}{k}$$

Forming a Team

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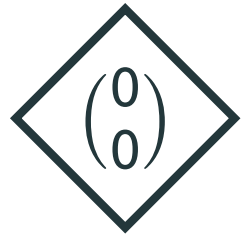
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- Fix one of the students, call her Alice
- There are two types of teams:
  1. Teams with Alice: $\binom{n-1}{k-1}$
  2. Teams without Alice: $\binom{n-1}{k}$
- Hence,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Pascal's Triangle

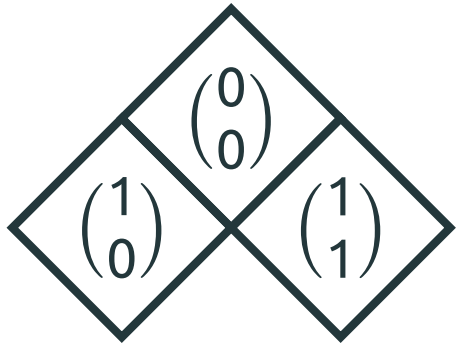
$n = 0$



Pascal's Triangle

$n = 0$

$n = 1$

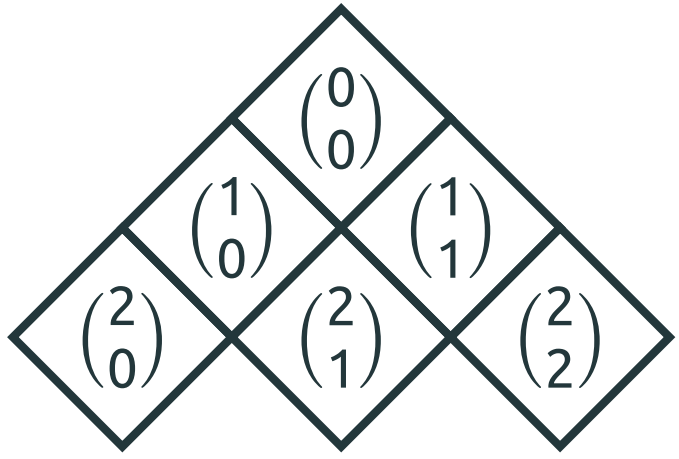


Pascal's Triangle

$n = 0$

$n = 1$

$n = 2$



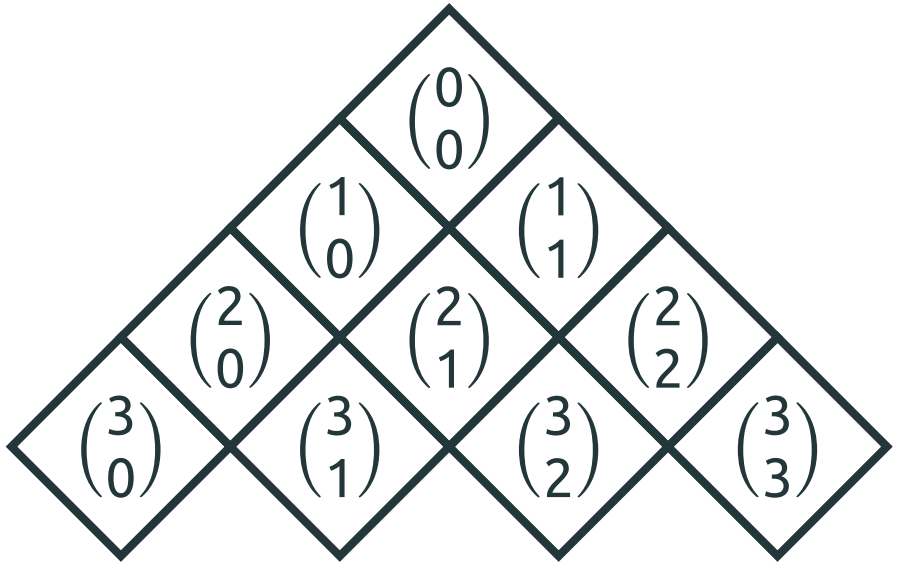
Pascal's Triangle

$n = 0$

$n = 1$

$n = 2$

$n = 3$



Pascal's Triangle

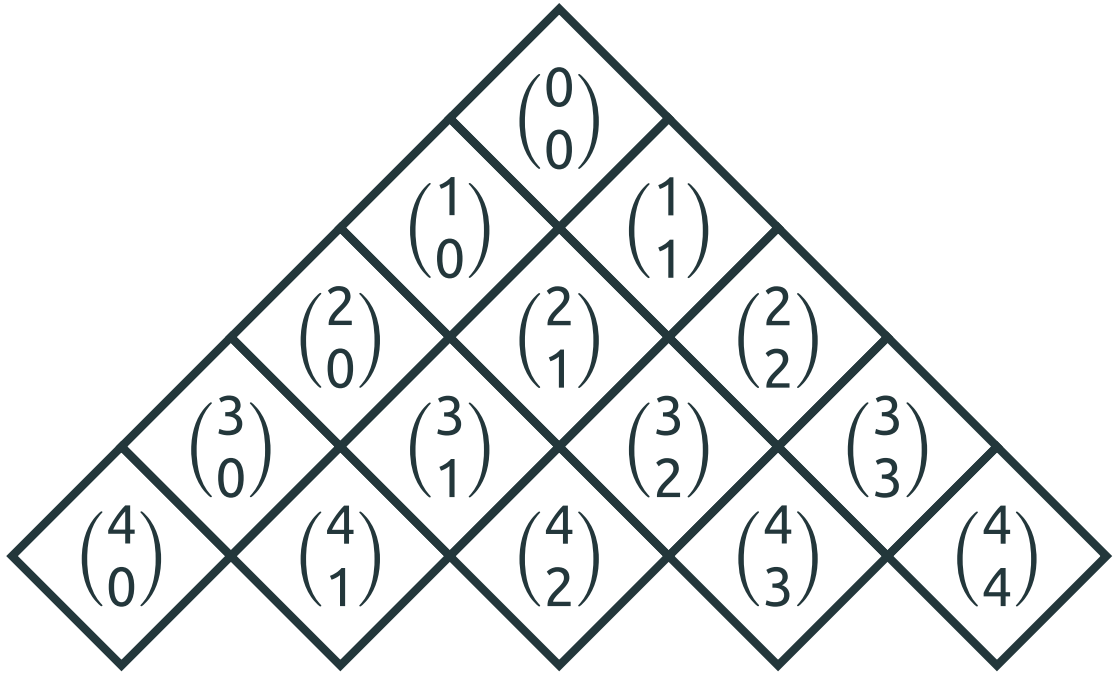
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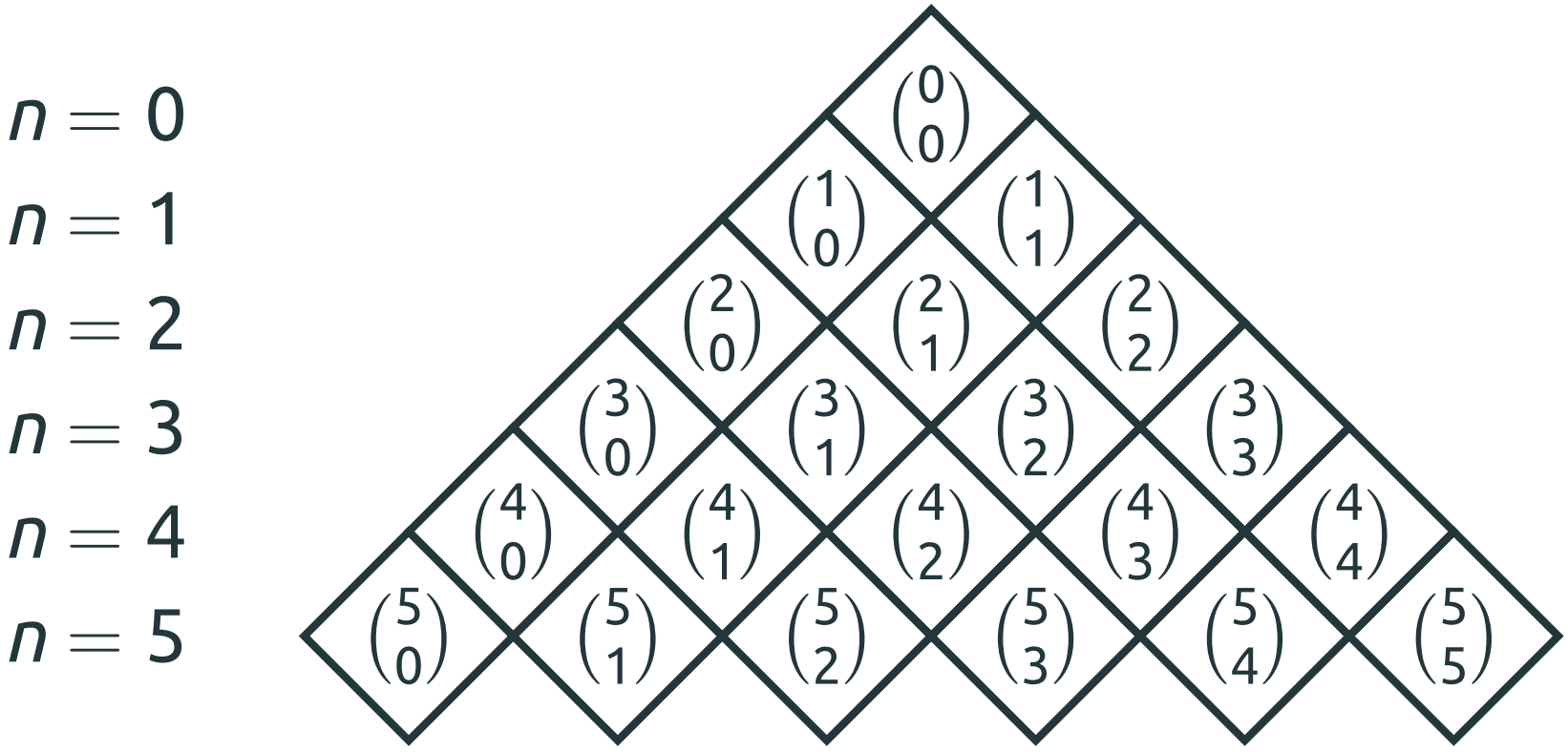
$n = 2$

$n = 3$

$n = 4$



Pascal's Triangle



Pascal's Triangle

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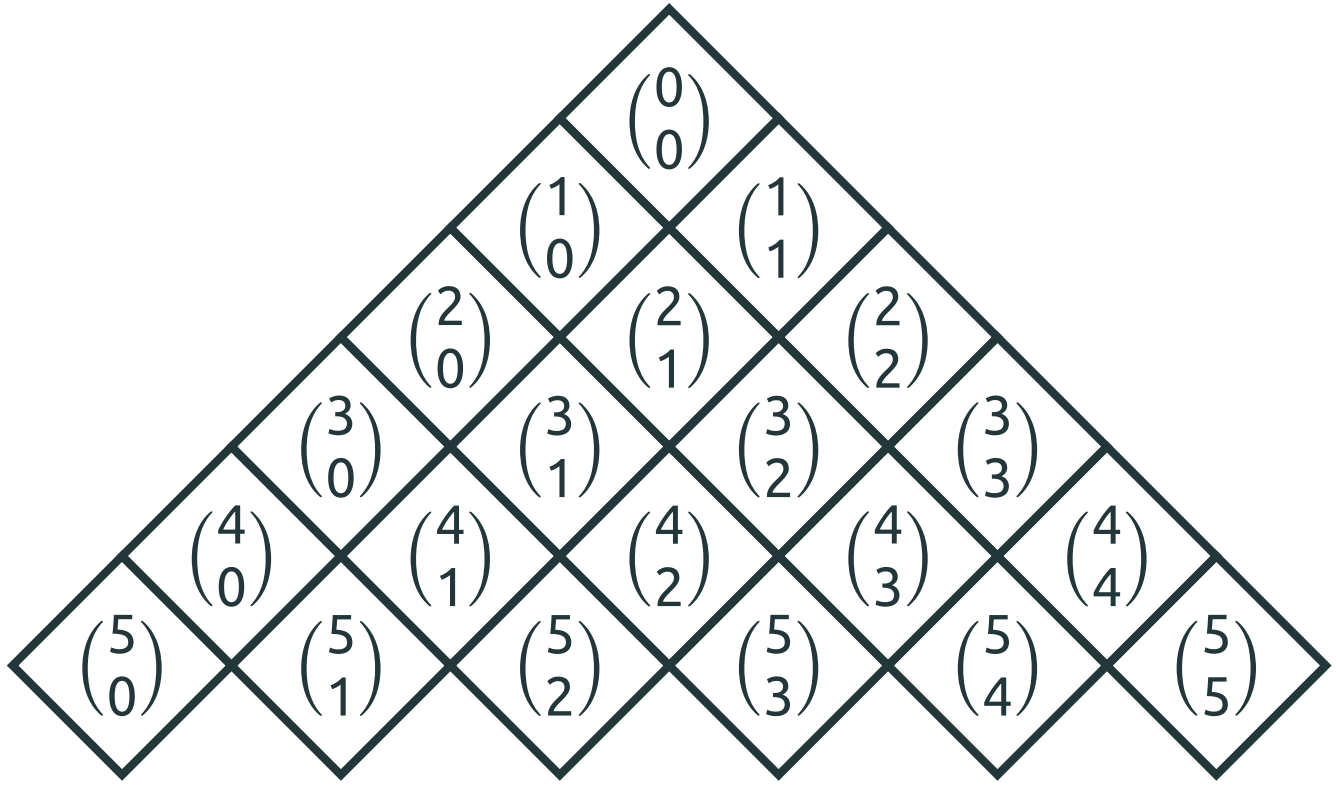
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$n = 4$

$n = 5$



$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

Pascal's Triangle

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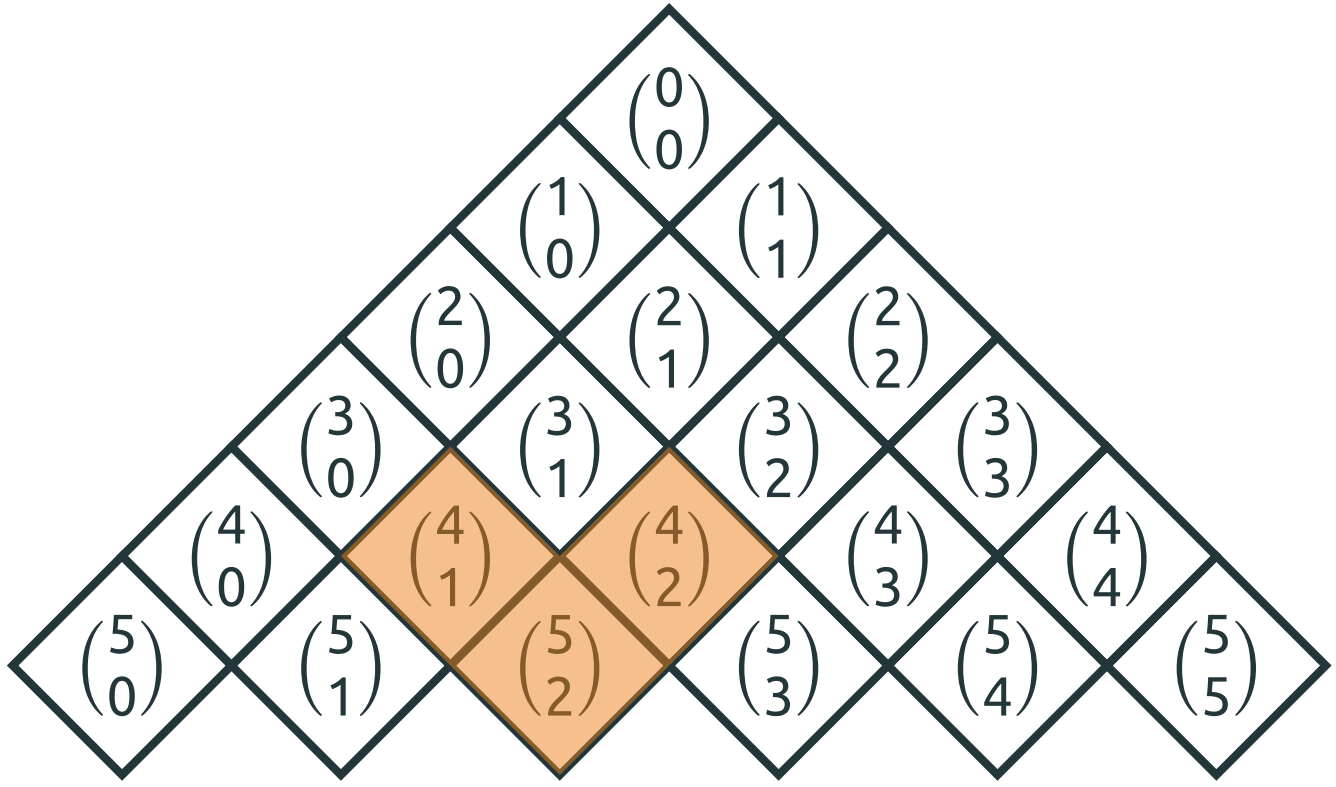
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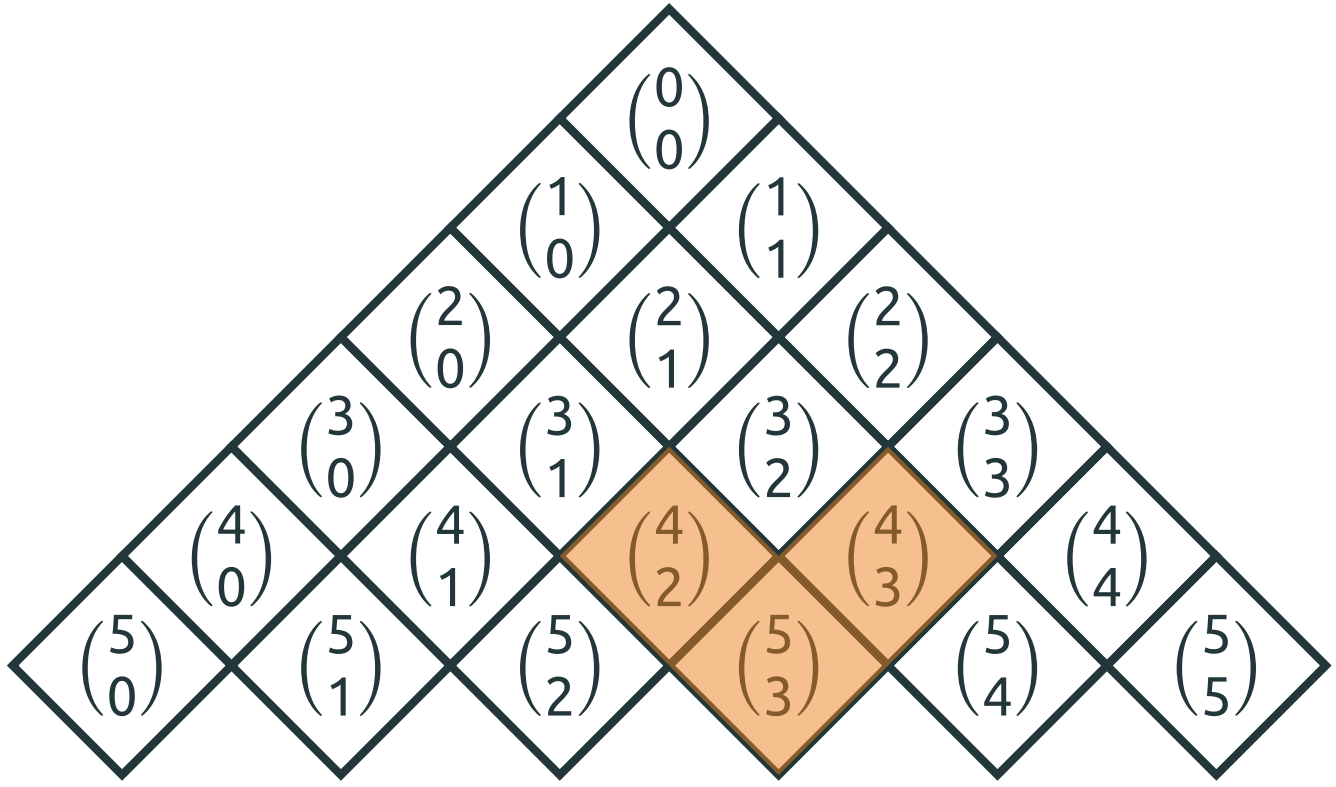
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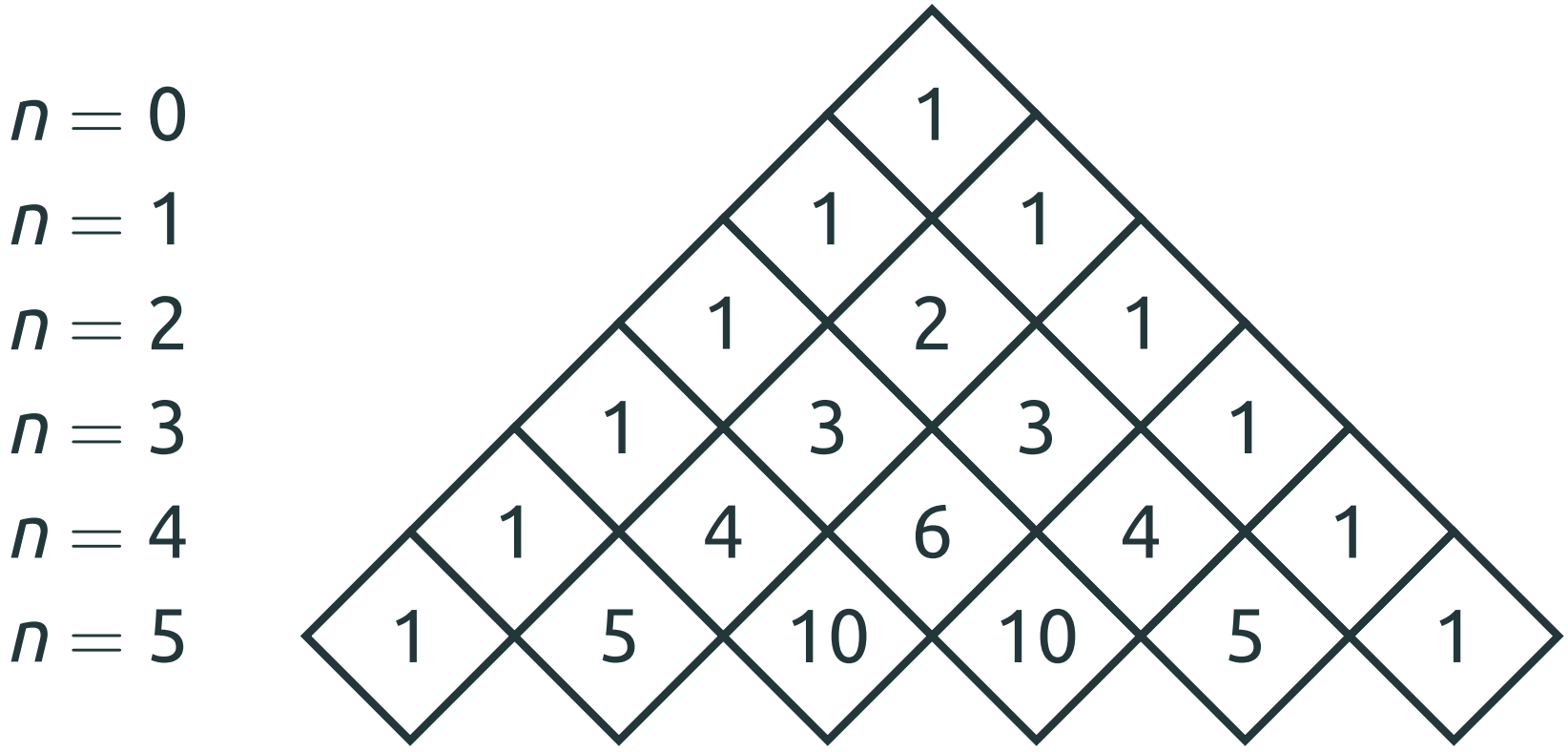
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Code

```
C = dict() # C([n,k]) is equal to n choose k

for n in range(8):
    C[n, 0] = 1
    C[n, n] = 1

    for k in range(1, n):
        C[n, k] = C[n - 1, k - 1] + C[n - 1, k]

print(C[7, 4])
```

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```
35
```

Outline

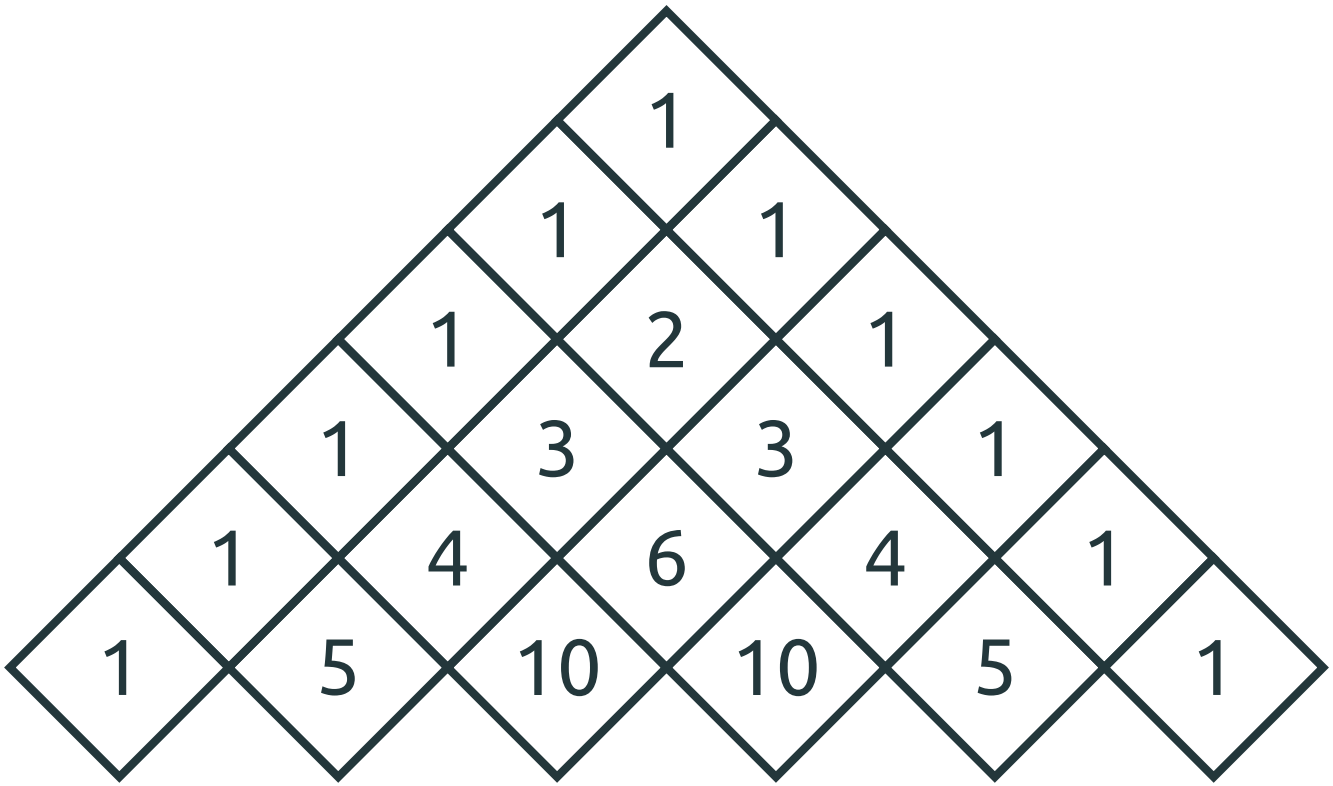
Pascal's Triangle

Symmetries

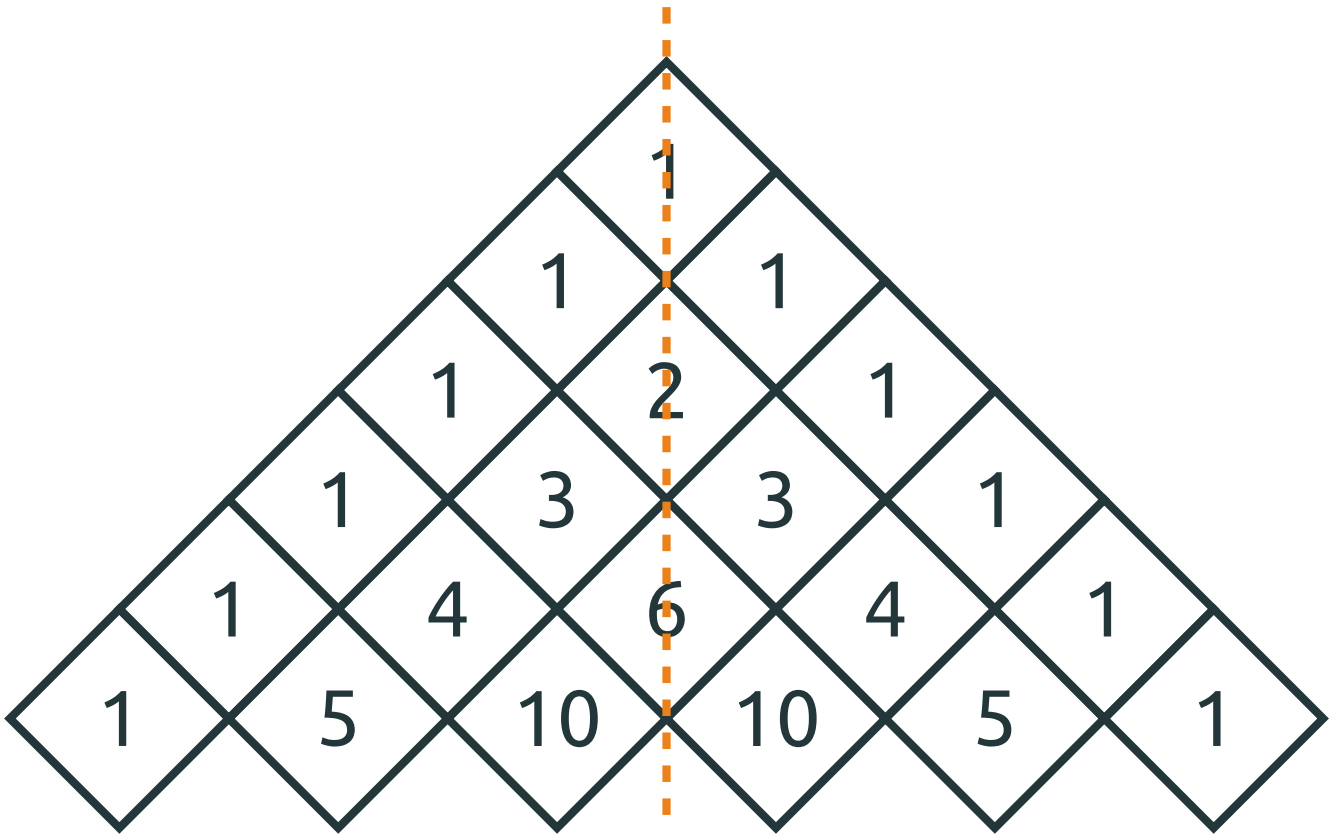
Row Sums

Binomial Theorem

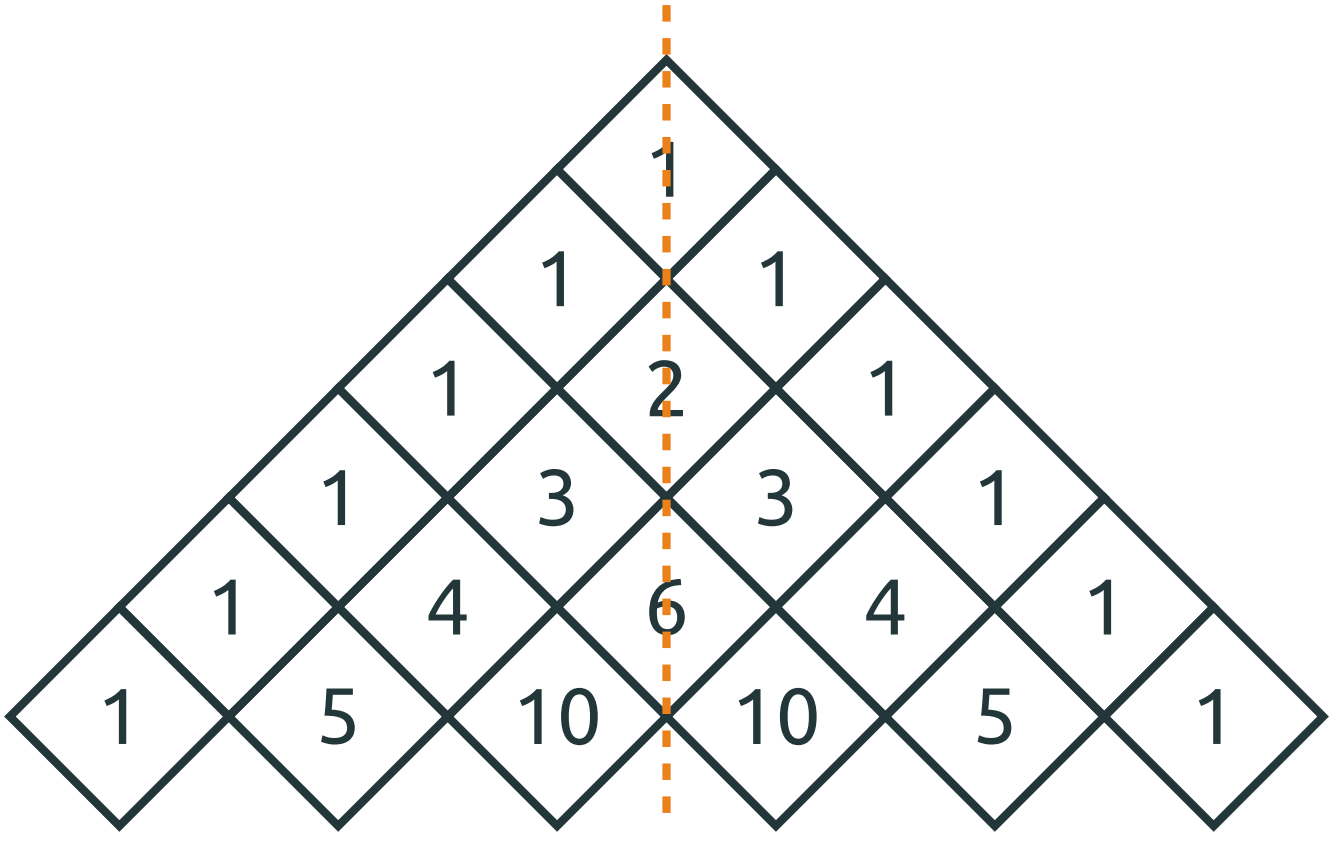
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$$\binom{n}{k} = \binom{n}{n-k}$$

Theorem

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Proof

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n!}{(n-k)!k!} = \binom{n}{n-k}$$

Combinatorial Proof

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- $\binom{n}{n-k}$ is the number of ways of selecting a team of size $n-k$ out of n students
- this is just the number of ways of partitioning n students into two teams of size k and $n-k$

Outline

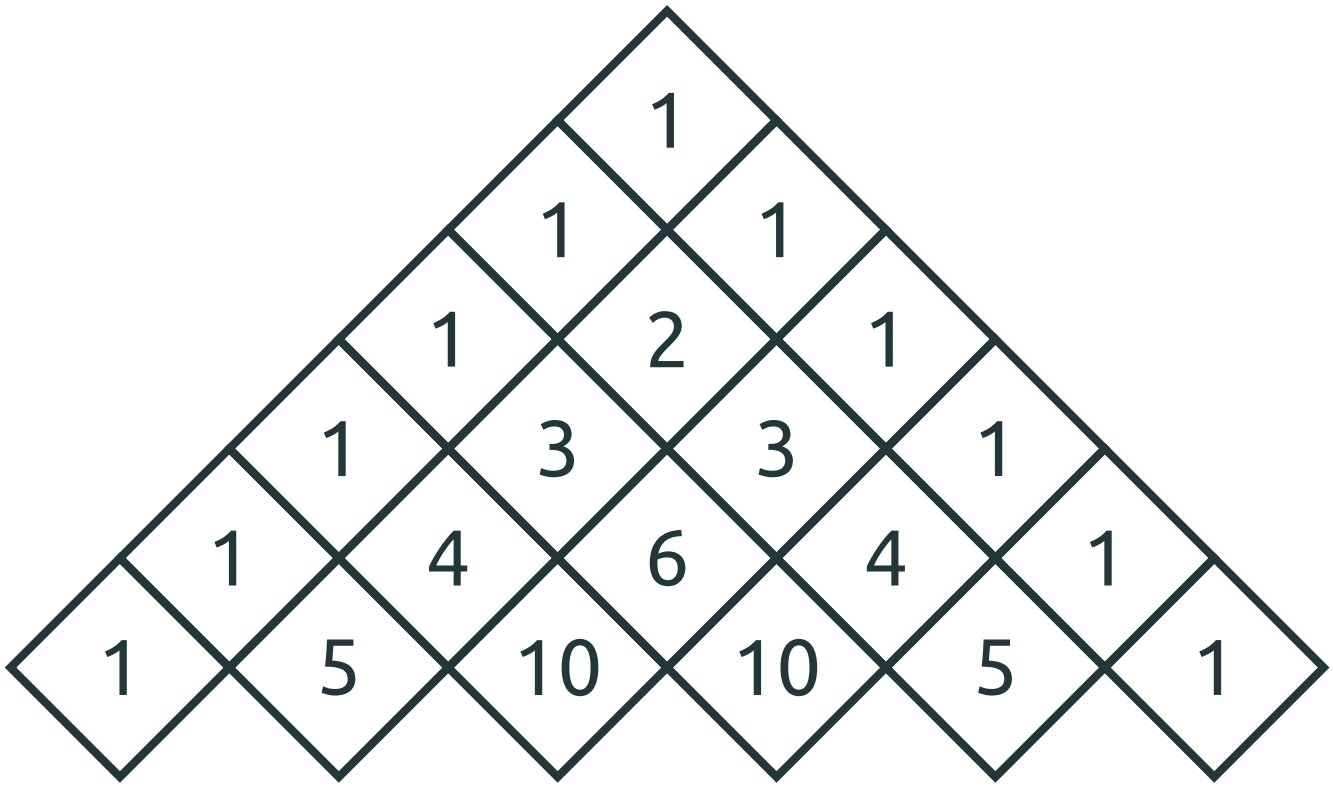
Pascal's Triangle

Symmetries

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Row Sums



Row Sums

$$1$$

$$1 + 1$$

$$1 + 2 + 1$$

$$1 + 3 + 3 + 1$$

$$1 + 4 + 6 + 4 + 1$$

$$1 + 5 + 10 + 10 + 5 + 1$$

Row Sums

$$
\begin{array}{ccccccccccc}
 & & & & 1 & & & & & & = 1 \\
 & & & 1 & + & 1 & & & & & = 2 \\
 & & 1 & + & 2 & + & 1 & & & & = 4 \\
 & 1 & + & 3 & + & 3 & + & 1 & & & = 8 \\
1 & + & 4 & + & 6 & + & 4 & + & 1 & & = 16 \\
1 & + & 5 & + & 10 & + & 10 & + & 5 & + & 1 = 32
\end{array}
$$

Theorem

The sum of all the numbers in the n-th row of Pascal's triangle is equal to 2^n:

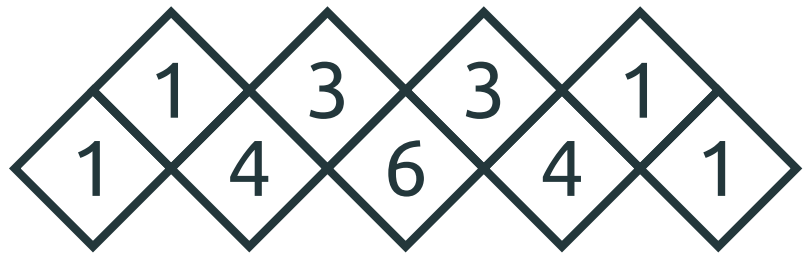
$$\binom{n}{0} + \binom{n}{1} + \cdots + \binom{n}{n-1} + \binom{n}{n} = 2^n$$

Proof by Induction

- The base case (0-th row) holds

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- We'll show that the sum of each row is twice the sum of the previous row:

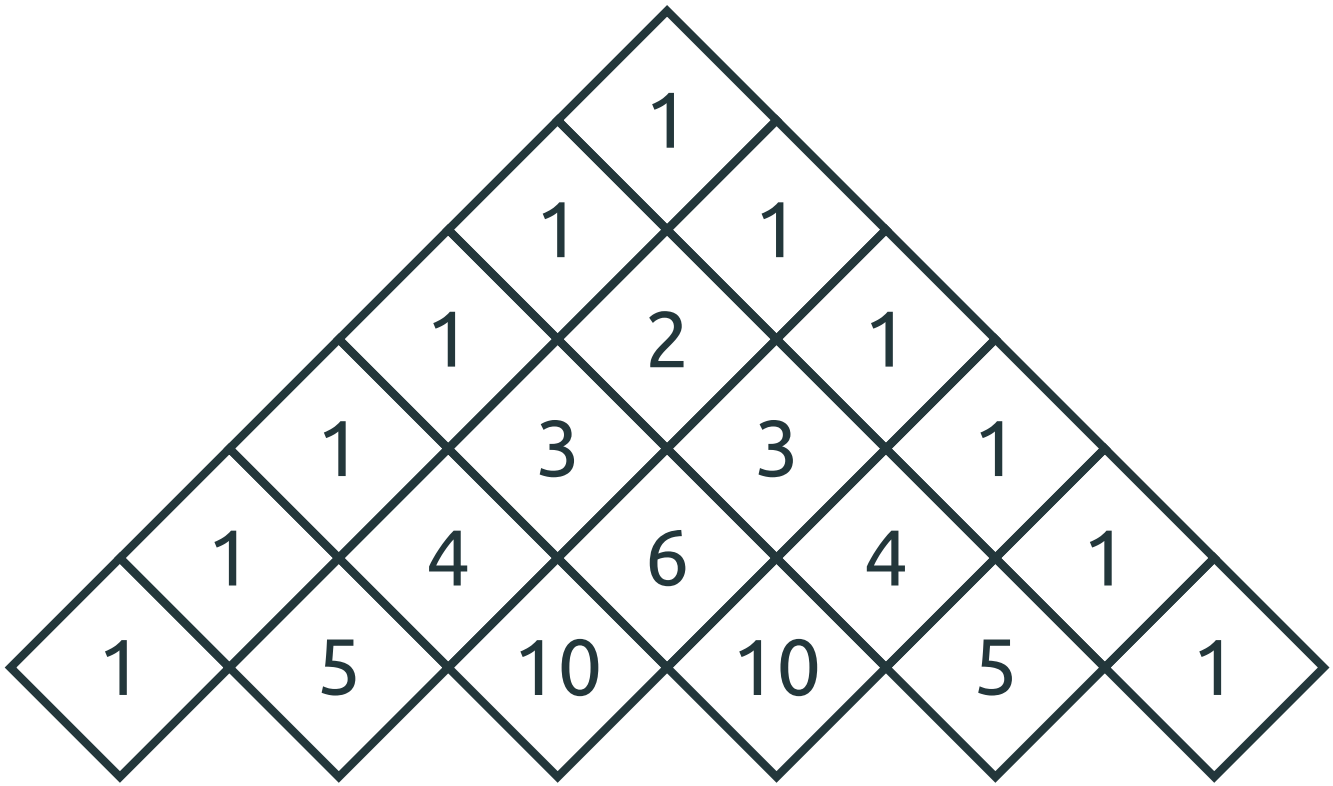


$$
\begin{array}{ccccccccc}
1 & & 4 & & 6 & & 4 & & 1 \\
1 & & (1+3) & & (3+3) & & (3+1) & & 1 \\
(1+1) & & (3+3) & & (3+3) & & (1+1) & &
\end{array}
$$

Combinatorial Proof

- $\binom{n}{k}$ is the number of k-subsets of a set of size n
- the sum of $\binom{n}{k}$ for all k (from 0 to n) is the number of all subsets of an n element set
- this is 2^n by the product rule: each of the n elements is either included or not

Alternating Row Sums



Alternating Row Sums

$$
\begin{array}{ccccccccccccc}
 & & & & & & 1 & & & & & & \\
 & & & & & 1 & - & 1 & & & & & \\
 & & & & 1 & - & 2 & + & 1 & & & & \\
 & & & 1 & - & 3 & + & 3 & - & 1 & & & \\
 & & 1 & - & 4 & + & 6 & - & 4 & + & 1 & & \\
 & 1 & - & 5 & + & 10 & - & 10 & + & 5 & - & 1 &
\end{array}
$$

Alternating Row Sums

$$
\begin{array}{ccccccccccccc}
 & & & & & & 1 & & & & & & \\
 & & & & 1 & - & 1 & & & & & = & 0 \\
 & & & 1 & - & 2 & + & 1 & & & & = & 0 \\
 & & 1 & - & 3 & + & 3 & - & 1 & & & = & 0 \\
 & 1 & - & 4 & + & 6 & - & 4 & + & 1 & & = & 0 \\
1 & - & 5 & + & 10 & - & 10 & + & 5 & - & 1 & = & 0
\end{array}
$$

Theorem

For $n > 0$, $\displaystyle\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0\,.$

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- For odd n, follows immediately from the symmetry property
- In general, can be shown by using the sum pattern of the triangle (each internal element is equal to the sum of the two elements above it)

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- Combinatorial meaning: the number of odd size subsets is the same as the number of even size subsets
- To prove this, we'll construct a one-to-one correspondence between odd size subsets and even size subsets

One-to-one Correspondence

- Fix any element x (can do this since $n > 0$)

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- Partition all subsets into pairs (A, B) where $A = B + x$ (more formally, $A = B \cup \{x\}$)
- One of A, B has odd size, the other one has even size

Example

$S = \{a, b, c, d\}$

Even size subsets	Odd size subsets
$\varnothing$	{a}
{a, b}	{b}
{a, c}	{c}
{a, d}	{d}
{b, c}	{a, b, c}
{b, d}	{a, b, d}
{c, d}	{a, c, d}
{a, b, c, d}	{b, c, d}

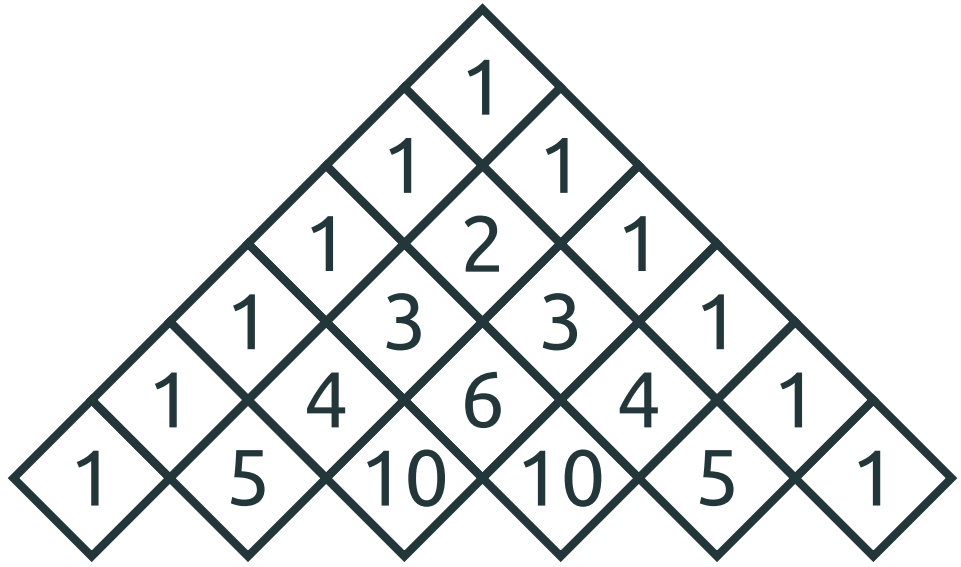
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Binomial Theorem



$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$$

$$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

Binomial Theorem

$$(a+b)^n = \binom{n}{0} a^n + \binom{n}{1} a^{n-1} b + \cdots + \binom{n}{k} a^{n-k} b^k + \cdots + \binom{n}{n} b^n$$

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Equivalently,

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k}b^k$$

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Equivalently,

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Can be shown by expanding the expression

$$(a+b)(a+b)\cdots(a+b)$$

Proof by Induction

$$
\begin{aligned}
&(a+b)^4 \\
&\qquad = (a+b)^3(a+b) \\
&\qquad = (a^3+3a^2b+3ab^2+b^3)(a+b) \\
&\qquad = a^4+3a^3b+3a^2b^2+ab^3+ \\
&\qquad\qquad +a^3b\ \ +3a^2b^2+3ab^3+b^4 \\
&\qquad = a^4+4a^3b+6a^2b^2+4ab^3+b^4
\end{aligned}
$$

Example

$$
\begin{aligned}
&(2a - b)^4 \\
&= ((2a) + (-b))^4 \\
&= (2a)^4 + 4(2a)^3(-b) + 6(2a)^2(-b)^2 + 4(2a)(-b)^3 + (-b)^4 \\
&= 16a^4 - 32a^3b + 24a^2b^2 - 8ab^3 + b^4
\end{aligned}
$$

Consequences

- Set $a = b = 1$. The number of subsets is 2^n:

$$2^n = \sum_{k=0}^{n} \binom{n}{k}$$

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$$2^n = \sum_{k=0}^{n} \binom{n}{k}$$

- Set $a = 1$, $b = -1$. The number of odd size subsets is the same as the number of even size subsets:

$$0 = \sum_{k=0}^{n} (-1)^k \binom{n}{k}$$

Another Consequence

- Set $a = 1$, $b = 2$:

$$3^n = \binom{n}{0} + \binom{n}{1}2 + \binom{n}{2}2^2 + \cdots + \binom{n}{n}2^n$$

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  - $\binom{n}{2}2^2$ is the number of words with exactly $n-2$ letters x